

PAPER_1333_DARK_AGE_21CM

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PAPER_1333 — Dark Age 21cm EDGES Signal

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Framework: UQFF — Star-Magic v5.27+

Tier: D — Astrophysics / Cosmology Open (CLOSED)

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Calculator: `calculate_paradox({"paradox": "dark_age_21cm"})`

Master Expression ``  $T_{21\text{cm}} = -D_{\text{phys}} \times A_5 \times \beta_i \times 2 = -289 \text{ mK}$  ``

****Match: between standard –200 mK and EDGES –500 mK****

Interpretation Deeper 21cm absorption signal via DPM buoyancy interaction at $z=17$.

UQFF Locked Primitives - $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$ - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT UQFF derives via integer-primitive identity. Zero fit parameters.

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